

Vision and Cognitive Systems - 25/06/24

4 questions, 90 minutes:

- 1) Describe an algorithm for edge detection (and define the kernel used). Report eventual similarities between this procedure and human vision.
- 2) Describe the Bag of Word pipeline. Propose a technique to address feature locality.
- 3) Describe the main components of a CNN (conv, pooling and FC layers). Report a typical architecture (take example from AlexNet).
- 4) You have this kernel: $16 \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ Define what is it and another kernel with which you can achieve a similar result.